Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A1EV · Group A

Report date 31 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**45**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A1EV-01** · Aneuploidy score: **18** · TMB (nonsyn., /Mb): **2.83** · MSIsensor score: **0.41** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

98th

High risk tier · fused neuro-symbolic model



Probability of progression

29% by 36 mo

12 mo 7% · 24 mo 17% · 60 mo 50%

Model-estimated progression time

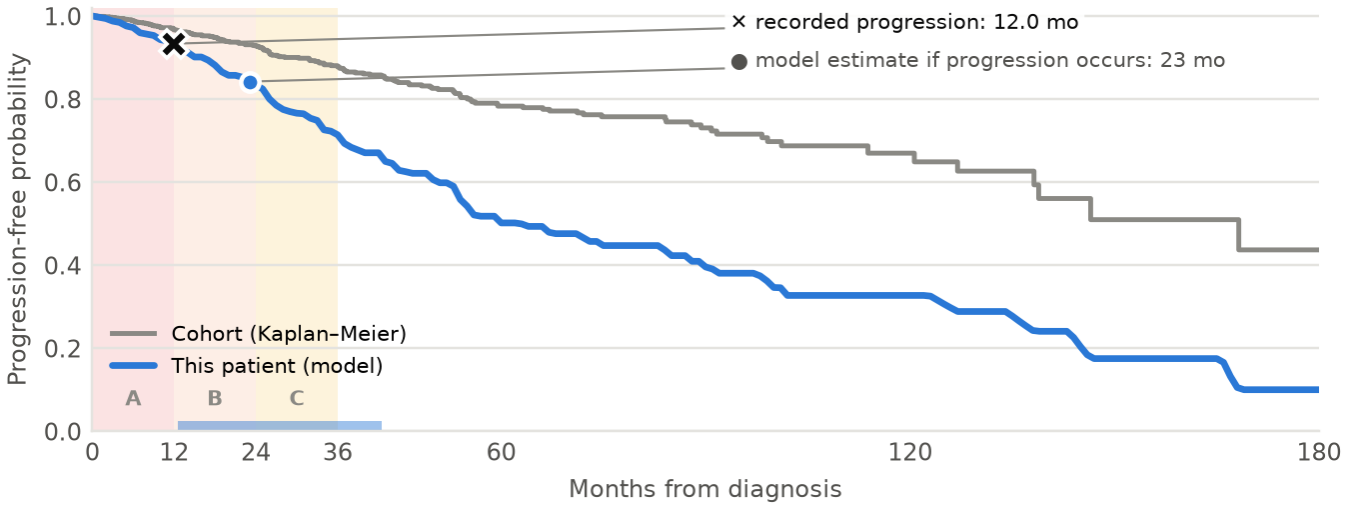
≈ 23 months

Window 12–24 months · IQR 13–42 mo
if progression occurs

Recorded progression

12.0 months

Group A · Early progression
✗ recorded window differs · |error| 11 mo



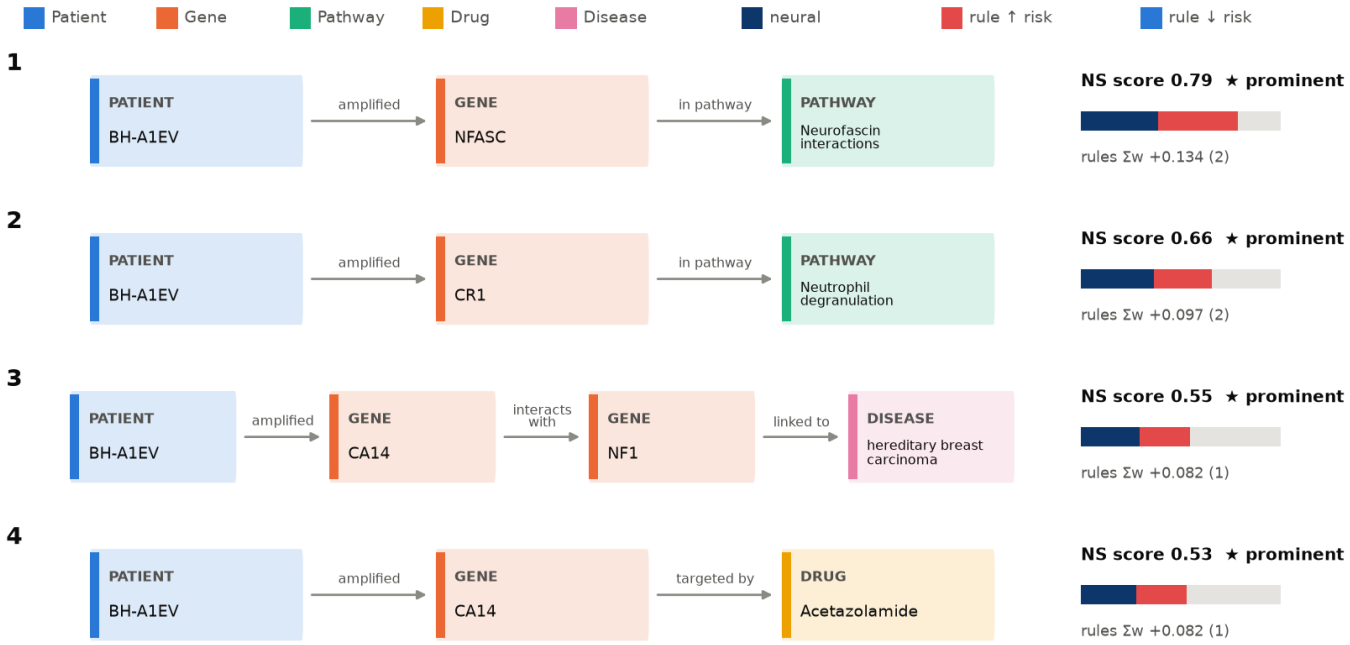
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 45-year-old patient has a Her2 tumour, AJCC stage III (T3, N1, M0). The fused neuro-symbolic model places them in the **98th risk percentile** (high tier). The strongest learned formula that holds for them (*Altered(p, g)* ∧ *TargetedBy(g, 'Ellagic acid')* ⇒ *Progress(p)*) raises the progression hazard (weight +0.082 per SD). The highest-scoring knowledge-graph path runs from **NFASC** to **Neurofascin interactions**; the learned formulas it instantiates point to faster progression (Σw +0.134). Progression was recorded at 12.0 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

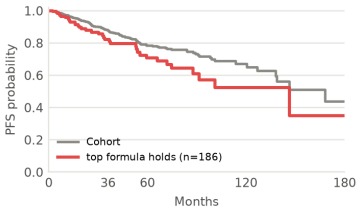


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A1EV** amplified → **NFASC** in pathway → **Neurofascin interactions**	64	↑ **faster progression** · Σw +0.134 over 2 formula(s) Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Neutrophil degranulation') ⇒ Progress(p) cohort: HR 1.51, n = 173 carriers	**0.79**
2 ★	**TCGA-BH-A1EV** amplified → **CR1** in pathway → **Neutrophil degranulation**	61	↑ **faster progression** · Σw +0.097 over 2 formula(s) Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Neutrophil degranulation') ⇒ Progress(p) cohort: HR 1.36, n = 316 carriers	**0.66**
3 ★	**TCGA-BH-A1EV** amplified → **CA14** interacts with → **NF1** linked to → **hereditary breast carcinoma**	49	↑ **faster progression** · Σw +0.082 over 1 formula(s) Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) cohort: HR 0.94, n = 792 carriers	**0.55**
4 ★	**TCGA-BH-A1EV** amplified → **CA14** targeted by → **Acetazolamide**	47	↑ **faster progression** · Σw +0.082 over 1 formula(s) Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) cohort: HR 1.73, n = 91 carriers	**0.53**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

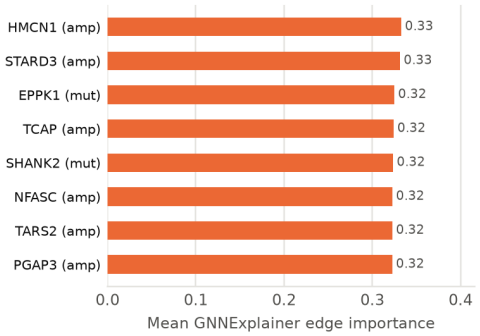
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.082 ↑ risk	Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) [KG]	single atom	185 pts
+0.079 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Neutrophil degranulation') ⇒ Progress(p) [KG]	mined pair	40 pts
+0.055 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Interaction between L1 and Ankyrins') ⇒ Progress(p) [KG]	mined pair	27 pts
+0.018 ↑ risk	Amplified(p, GOLPH3L) ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Complement cascade') ⇒ Progress(p) [KG]	mined pair	42 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
BRCA2 · driver	High-level amplification	GISTIC ±2	14	0.31
HMCN1	High-level amplification	GISTIC ±2	0	0.33
STARD3	High-level amplification	GISTIC ±2	1	0.33
EPPK1	Mutation	p.L1474=	0	0.32
TCAP	High-level amplification	GISTIC ±2	1	0.32
SHANK2	Mutation	p.P1240T	1	0.32
NFASC	High-level amplification	GISTIC ±2	3	0.32
TARS2	High-level amplification	GISTIC ±2	2	0.32
PGAP3	High-level amplification	GISTIC ±2	0	0.32
MIR4728	High-level amplification	GISTIC ±2	0	0.32
ZPBP2	High-level amplification	GISTIC ±2	0	0.32



46 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 36 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-C8-A1HL	38	III	LumB	0.70	no progression, followed 10.4 mo
TCGA-A8-A07I	69	III	Her2	0.66	no progression, followed 14.0 mo
TCGA-B6-A0IQ	40	III	Basal	0.62	no progression, followed 140.9 mo
TCGA-B6-A0WX	40	III	Basal	0.62	progressed at 18.5 mo (group B)
TCGA-A8-A09B	58	III	LumA	0.62	no progression, followed 12.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.